###
Student Name: Rong Wei
Course Code: CSCI 460 -- Fall 2026
Assignment Due Date: 10/25/2026
GitHub Link:
https://github.com/RMarioOli/CSCI460F2026_Programming_Assignment1_W30779947
###

Effect of Training Data Size on Naive Bayes Classification Performance

The purpose of this experiment was to investigate how the amount of training data affects the classification performance of a Gaussian Naive Bayes model. The model was evaluated using two performance metrics: accuracy and weighted F1-score.

The training data percentage was increased from 10% to 90% in increments of 10%. For each training data percentage, the dataset was randomly split into training and testing sets using stratified sampling to maintain approximately the same class distribution as the original dataset. The model was trained on the training set and evaluated on the testing set. Each experiment was repeated 10 times with different random seeds to examine the consistency of the results

For each training data percentage, the average accuracy, standard deviation of accuracy, average weighted F1-score, and standard deviation of the weighted F1-score were calculated across the 10 trials. The mean values were used to compare overall classification performance, while the standard deviations were used to evaluate the variability of the results across trials.

The results show that increasing the training data generally improved the model's classification performance, although the improvement was not consistent at every step. When the training data increased from 10% to 40%, average accuracy remained nearly unchanged, ranging from 0.8388 to 0.8394. The average weighted F1-score also remained stable, between 0.8475 and 0.8481. This suggests that adding training data within this range had little impact on the model's overall performance.

A more noticeable improvement occurred as the training data increased from 40% to 60%. Average accuracy increased from 0.8394 to 0.8428, and the average weighted F1-score increased from 0.8481 to 0.8503. These results suggest that the model benefited from having more observations available to estimate the class distributions.

However, performance did not increase continuously. At 70% training data, average accuracy decreased to 0.8409, and the average weighted F1-score decreased to 0.8488. Both metrics increased again at 80%, and the highest average scores occurred at 90% training data, with an accuracy of 0.8495 and a weighted F1-score of 0.8552. These fluctuations may result from differences in the random training and testing splits rather than a negative effect of additional training data.

Importantly, a higher training percentage does not automatically guarantee better generalization. The purpose of training data is to help the model learn patterns that also apply to unseen observations. Test performance is therefore essential when evaluating whether the additional training data actually improves generalization. Overfitting is typically associated with training performance being substantially better than test performance, while underfitting is associated with poor performance on both training and testing data.

The standard deviations also changed as the training data increased. The standard deviation of accuracy rose from 0.0042 at 10% training data to 0.0111 at 90%. Similarly, the

standard deviation of the weighted F1-score increased from 0.0034 to 0.0068. This indicates that the results became more variable across trials at higher training percentages.

One possible explanation is that increasing the training percentage reduces the size of the testing set. At 90% training data, only 10% of the observations remain for testing, making the evaluation more sensitive to the particular observations selected. Consequently, the highest average scores at 90% should be interpreted with caution because they were accompanied by the largest standard deviations.

Another limitation concerns the assumptions of the Gaussian Naive Bayes model. It assumes that features are conditionally independent given the target class and that numerical features follow Gaussian distributions. However, customer characteristics in the Bank Marketing dataset may be related to one another, and not all numerical features necessarily follow Gaussian distributions. These assumptions may limit the model's ability to capture more complex relationships in the data.

Overall, increasing the training data from 10% to 90% was associated with a modest improvement in the Gaussian Naive Bayes model's test performance, with the largest average accuracy and weighted F1-score occurring at 90%. However, the nearly unchanged results at lower training percentages and the fluctuations at 70% and 80% show that the relationship was not strictly linear.

In conclusion, using more training data generally benefited the model in this experiment, but the highest average scores did not necessarily represent the most stable estimates of performance. Further evaluation using repeated stratified cross-validation or a separate, fixed test set would help determine whether the observed improvements generalize to unseen data. The results suggest that additional training data can help the model learn more representative patterns, but the size and composition of the testing set also affect the measured performance.